

Differentiable functions – 6

Differential inequalities and comparison

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Gronwall). If u is continuous, non-negative and $u(t) \leq a + \int_{t_0}^t b(s)u(s) ds$ with $b \geq 0$, then $u(t) \leq a \exp(\int_{t_0}^t b)$. The differential form: $u' \leq bu$ implies $u(t) \leq u(t_0) \exp(\int b)$; the proof is one line after multiplying by the integrating factor.

Theorem 2 (comparison). If $u' \leq f(t, u)$, $v' = f(t, v)$ and $u(t_0) \leq v(t_0)$ with f locally Lipschitz in the second argument, then $u \leq v$ for $t \geq t_0$: a differential inequality is inherited by the solutions.

Theorem 3 (Sturm). Between two consecutive zeros of a nontrivial solution of $y'' + q_1 y = 0$ there is a zero of every solution of $y'' + q_2 y = 0$ whenever $q_2 \geq q_1$; in particular a solution of $y'' + qy = 0$ with $q \geq m^2 > 0$ has a zero in every interval of length $\frac{\pi}{m}$.

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1. Prove Gronwall's lemma in both forms, and use it to prove uniqueness for the Cauchy problem $y' = f(t, y)$, $y(t_0) = y_0$ with f Lipschitz in y .
 2. Prove the comparison theorem and deduce: if $f'(x) \leq f(x)$ for all $x \geq 0$ and $f(0) = 1$, then $f(x) \leq e^x$; give an example showing that the inequality may be strict for all $x > 0$.
 3. Prove Sturm's comparison theorem and deduce that every solution of $y'' + q(x)y = 0$ with $q(x) \geq 1$ has infinitely many zeros on $[0, \infty)$, while for $q(x) = \frac{1}{4x^2}$ it need not.
 4. Let f be twice differentiable on $[0, \infty)$ with $f''(x) \geq f(x)$ and $f(0) = f'(0) = 0$. Prove that $f \equiv 0$; deduce the corresponding uniqueness statement for the equation $y'' = y$.

5. Prove that the Riccati substitution $u = \frac{y'}{y}$ turns $y'' + q(x)y = 0$ into $u' + u^2 + q = 0$, and use it to show that a solution of $y'' + qy = 0$ with $q > 0$ and $\int^\infty q = \infty$ cannot be eventually positive.